Week 6 web page: delivery guide

Instructor notes for Week6_LikelihoodAndPowerAnalysis.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week6_LikelihoodAndPowerAnalysis.html. Open it before class; the widgets need no internet once loaded.
- For live demos, run the matching section of `CODE_Week6_LikelihoodAndPowerAnalysis.R` (same seeds, same numbers). The power grid takes 20-30 s; on a slow laptop use `R = 200` live and say why (rough first, more replicates only near the threshold).
- Zoom the browser to 125-150% for the projector.
- Questions on the page have no answers. Answers are below; *Land:* is where discussion should end up.
- Readings for Canvas: Bolker 6.1-6.2.1 and 6.3.1 (printed pp. 227-236, 243-247); 5.3 (pp. 209-226) for the power analysis.
- WODs: none Monday (the period opened with a debrief of the previous Wednesday's *Writing the Likelihood* WOD, done while the instructor was away). Wednesday: *Find the Bugs* (two bugs, ~5 min).

As taught (Sept. 28-30, 2026)

- **Monday:** debrief of *Writing the Likelihood* (focus on 2d, 3b, 4c, 5c), then a ~20-min course check-in: AI disclosure (AI drafts handouts, keys, and notes; the course design, editing, and checking are the instructor's), and anonymous index cards (side 1: start / stop / continue on the class format; side 2: the Bolker book, what's helpful, frustrating, or confusing, and how much of the reading students do). Labs dropped to two shorter Acts from this week. The last ~15 min was the likelihood and MLE overview (page sections 3-4, with the pika example). Per the announcement (added Nov. 4): the logistic regression formula was written in class without the minus sign in the exponent; the announcement corrected it, and the Week 7 page's `#logistic` now gives the binomial likelihood with the sign flagged. The AI handout process is now described in the Week 1 page's `#course`; feedback came by anonymous WebCampus poll.
- **Lab 5:** about the right length, but only one group finished both Acts. Students asked Monday for lab topics in advance; the instructor's pre-lab review list (Bolker 6.1-6.2.1, the binomial calculus MLE, the numerics, the myxomatosis gamma fit, custom functions, binomial/Poisson, monomolecular) is now at the top of `#lab5`. Lab 5 was likely the course's one by-hand calculus MLE. Going forward the instructor aims to post lab prep by the end of the prior week. Per the post-lab announcement: the opening covered poll

takeaways, the big picture for likelihood and MLE, the Poisson MLE by calculus on the board ($\hat{\lambda} = \bar{y}$; now written out in `#mle`), and the same problem in R with `dpois()`, a likelihood function, and an NLL function. Act 2 went home, due the next Monday.

- **Wednesday:** the two-bug WOD, then a live-coding walkthrough of the Lab 4 monitoring design problem (never handed out as a handout). The AI-prompting part was skipped; the page keeps it as a bonus. The demo didn't finish; the instructor completed the code after class and posted it (lightly commented; commenting it was suggested as an exercise). The announcement's view of where AI helps in a power analysis (visualizing power over many designs, stronger optimization with more design variables, a simple web app; not the core functions) now opens `#ai-prompts`. It also moved Bayesian inference earlier, following Bolker (framing now in the Week 7 page's `#bayes-big-picture`), and previewed Lab 6 (2-D surfaces, quadratic approximation; prep note now atop `#lab6`).
- The page was built after the week from the two PDF NOTES (now in `lecture_notes/archive/`) and the legacy LECTURE4 page (now in `archive/`). The tree example, the deviance bonus, and both widgets are new; the tree bonus replaces LECTURE4's mtcars demo.

Part 1: Likelihood and MLE (Monday, ~35 min if taught in full)

Big picture (~2 min)

- Point at FIT. Forward (simulate data from parameters) vs. backward (data fixed; which parameters?).

WOD: writing the likelihood (~10 min)

- Bernoulli formula, product under independence, $p^3(1-p)$.
- "Add up the three likelihood values. Do they sum to 1? Should they?" *Land:* $46/256 \approx 0.18$, and the area under the whole curve is only $1/20$. Probabilities sum to 1 over *outcomes* with p fixed; a likelihood runs over *parameter values* with the data fixed, and nothing makes those add to 1. $\mathcal{L}_4(0.75)$ is not "the probability that $p = 0.75$."
- Also worth a minute from the WOD: 4c (the covariate model is twice as likely, but nobody searched for the best a, b; that search is MLE).

From likelihood to NLL (~6 min)

- Underflow: $0.5^{1852} \approx 10^{-558}$ rounds to 0; the sum of logs is -1283.7. Minus sign: optimizers minimize, nothing deeper.
- Widget (default data, slide λ):
  - $\lambda = 1.5$: probabilities 0.251, 0.014, 0.126, 0.223, 0.047, 0.047; likelihood 2.2×10^{-7}, NLL 15.33.
  - $\lambda = 3$: likelihood 7.1×10^{-6}, NLL 11.85 (the minimum).
  - $\lambda = 6$: the zero count has probability 0.0025; NLL 17.38.

  - "Four times as many plots," $\lambda = 3$: likelihood 2.6×10^{-21}, NLL 47.41, and the curve is much steeper.
- "Slide λ from 1.5 to 3 to 6. Which observed count is most to blame for the poor likelihood at each end? Then switch to 'Four times as many plots.' What happens to the likelihood at the MLE, and what happens to the shape of the NLL curve?" *Land:* at 1.5 the count of 5 (0.014); at 6 the zero (0.0025). With four times the data the likelihood at the MLE collapses (7×10^{-6} to 3×10^{-21}) but the NLL valley gets four times steeper. Height means nothing; shape is information (next week's intervals).

The MLE (~5 min)

- Same location on all three scales (log is monotonic). Don't derive $\hat{\lambda} = \bar{y}$ on the board if the lab hasn't happened yet: Lab 5 Act 1 asks for the analogous binomial derivation.
- Three routes; Brent for one parameter.

Discussion (~5 min)

- Q1 (likelihood 7.1×10^{-6}, "the model must fit badly") *Land:* no. The height shrinks with every observation; it only compares parameter values (or models) on the same data. To judge fit, simulate from the fitted model, compare against alternatives, or use the deviance (page bonuses).
- Q2 (independent plots?) *Land:* neighboring plots share talus quality, forage, and possibly the same pikas, so counts are correlated. The MLE may barely change, but the likelihood treats six correlated plots as six independent pieces of information, so the curve is too steep and we're overconfident.

Bonus: fitting by simulation (LECTURE4's demo with trees)

- "Why not simply increase σ until every point falls inside the band?" *Land:* a wide enough band "fits" anything and says nothing; the likelihood penalizes it, since spreading density thin lowers every observation's density (NLL 114.5 for the wide band vs. 83.9 for the first guess).

Part 2: Fitting a model with several parameters (Lab 5 opening)

- Examples only with the tree data; never the lab's frog numbers. Don't show before the Acts: the binomial derivative (Act 1 Part 2), that constants don't move the peak (Act 1 1c), or why `optim()` prints NaN warnings (Act 2 2b).
- Density vs. probability: the first tree is 0.124 per cubic foot but 4.38 per cubic meter; the shift is the same for every parameter value, so the MLE doesn't move.
- NLL function anatomy (four bullets), starting values from meaning ($b \approx 2$, $a \approx 10.3/8.3^2 = 0.15$, $\sigma \approx 3$), `optim()` gives $a = 0.087$, $b = 2.24$, $\sigma = 3.18$, NLL 79.86; a second start agrees.

- "The model assumes the same noise (σ) for small and large trees. Does the plot support that? If not, what kind of distribution might describe volume better?" *Land:* the scatter grows with size (small trees hug the curve; the band is too wide there). A gamma or lognormal (spread proportional to the mean) fits the biology better.
- "An exponential curve has its best NLL at 81.57, compared with 79.86 for the power law. Is a difference that size enough?" *Land:* same number of parameters, so a lower NLL is a fair comparison of fit, but 1.7 units isn't decisive on its own; check residuals and the biology (geometry predicts a power law). Formal tools (AIC, LRT) come later.
- Act keys: `labs/answer_keys/Lab5_Act1_Key.pdf`, `labs/answer_keys/Lab5_Act2_Key.pdf`.

Part 3: A power analysis in code (Wednesday, ~40 min)

WOD (~5 min, runs ~10)

- "Find each bug. For one of them, what quick check would have caught it?" *Land:* `1 - pobs^k` should be `1 - (1 - pobs)^k` (check `detect_prob(1)` equals 0.02); `detected <-` should be `detected[i] <-` (check the result is near $1 - 0.6^3 = 0.784$, never exactly 0 or 1). Both run without errors: "it runs" isn't a test.

Pieces (~15 min)

- Type the four functions live; say each check before running it (`detect_prob(1)`; a vector of two years; `length()` is 25; the stable-population run gives 0.023, about half of 5%).
- Mention two more bugs here: `rbinom(1, ...)` in `simulate_counts()` (one count, not 25) and `||` for `&&` in `detect_decline()` (about 50% false declines).

Power across designs (~15 min)

- False-decline rate 0.025, power 0.822 at $k = 3$, $s = 1$.
- Cheapest passing designs: $k = 10$, $s = 4$, \$28,000, power 0.750; then \$29,400 (0.834) and \$30,600 (0.753). Re-check with 5,000: 0.757, 0.805, 0.738 (SE about 0.006); the third drops below 0.75.
- Widget: default $k = 3$, $s = 1$ (25 surveys, \$65,000, 0.827). Then $k = 10$, $s = 4$ (7 surveys, \$28,000, 0.750).
- "Click 'Simulate another program' ten times. How many detected the decline? Why is any single program so unpredictable, even when the power is 0.75?" *Land:* expect about 7 or 8, but anything from 5 to 10 is common (binomial with $n = 10$, $p = 0.75$). Power is a long-run rate; one program is one draw, and with only 7 surveys, detection noise can hide or exaggerate the trend.

Discussion (~5 min)

- Q1 (cheapest at 0.757 vs. \$29,400 at 0.805) *Land:* $k = 11$, $s = 4$: \$1,400 more, and it clears the goal by about 10 Monte Carlo SEs instead of 1. The model is also optimistic (next question), so margin matters.

- Q2 (a check to add) *Land:* any known-answer case, e.g. a deterministic declining series in `detect_decline()` returns `TRUE`; the mean of many simulated counts equals $N_t q(k)$; `cost_of(1, 25)` is one survey.
- Q3 (good and bad years that carry over) *Land:* persistent environmental noise makes consecutive counts correlated, which the regression ignores. Power drops and false declines rise: the Lab 4 Act 2 key's version (log-scale SD 0.05) gives power about 0.50 and false declines about 15% for $k = 11$, $s = 4$.

Bonus: prompting an AI assistant (not covered; available for a later AI live-coding demo)